

LYCEUM OF THE PHILIPPINES UNIVERSITY – LAGUNA

College of Business and Accountancy

MIDTERM ASSESSMENT — TEST II PRACTICAL

Bayanihan Bus Lines · 45 points · 120 minutes

One task per sheet. Write your EVIDENCE and DECISION here and hand it in.

Name: Lagdamen, Lawrence M. Drawn at: 10:23AM Finished: 10:33AM

BAND A · TASK A1 · 8 POINTS

A single operation with one correct outcome

What did you just load?

1. Load BayanihanBus_Tickets.xlsx.
2. Report the number of rows and the number of columns.
3. Count the DISTINCT values of TicketID and compare it with the row count.

EVIDENCE

Rows, columns, distinct TicketID, and the difference between the last two.

No. of Rows: 18329 rows

No. of Columns: 7 columns

Every distinct value of TicketID is equivalent to the number of rows. Meaning, that for every row of the Tickets file has a distinct value of TicketID for every ticket purchased.

DECISION

Written, in full sentences.

Every row in the BayanihanBus_Tickets.xlsx file represents every ticket sold on a certain trip, on certain Route, on a particular day.

Applied steps renamed? **YES** / NO Query name: Data type changed to "Date"

Name: Lagdamen, Lawrence M. Drawn at:10:33AM Finished: 10:43AM

BAND A · TASK A2 · 8 POINTS

A single operation with one correct outcome

The column that cannot be used

4. Load BayanihanBus_Trips.xlsx.
5. Report the number of rows.
6. Report the data type Power Query assigns to OdometerKM.
7. State in one sentence why that type is a problem.

EVIDENCE

Row count, the assigned type, and the consequence.

No. of Rows: 1200 rows

OdometerKM data type: Whole number

The OdometerKM column should be in decimal number data type format, because some of these buses have decimal readings on the Odometer which represents "Meter".

DECISION

Written, in full sentences.

ESCALATE, This column alone lacks supporting data like total distance traveled which would complement the column "OdometerKM", this data alone only represents the current Odometer reading of the bus unit after the certain trip has been completed.

Applied steps renamed? **YES** / NO Query name: Data type changed to "Date"

Name: Lagdamen, Lawrence M. Drawn at: 10:43AM Finished: 10:55AM

BAND A · TASK A3 · 8 POINTS

A single operation with one correct outcome

How many tickets to a trip?

8. Load both Tickets and Trips.
9. Group the tickets by TripID and count the rows in each group.
10. Report the minimum, the maximum and the mean.

EVIDENCE

Min, max and mean tickets per trip — and one sentence on what this proves.

Tickets:

Min – 6

Max – 26

Mean - 16

DECISION

Written, in full sentences.

The numbers presented above are the number of trips that happened for each TripID. With one TripID only having a total of 6 trips, and another TripID having the most trips of 26. Together these data represent a mean of 16 trips.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M. Drawn at: 10:55AM Finished: 11:08AM

BAND B · TASK B1 · 10 POINTS

Two steps where the order matters

Make the odometer usable

11. OdometerKM will not convert to a whole number. Repair it.
12. Report how many values contained a comma.
13. Report the minimum and maximum odometer reading after the repair.

EVIDENCE

The comma count, the min and max, and the order of your two steps.

Rows with comma: 15 rows

Min: 48000

Max: 611741

DECISION

Written, in full sentences.

Since OdometerKM is not being properly read because of a formatting error, I first changed the data type into "Text" and then replaced the values containing a comma, then I changed it back to Whole number, and did my analysis based on the fixed values presented.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M Drawn at: 11:08AM Finished: 11:16AM

BAND B · TASK B2 · 10 POINTS

Two steps where the order matters

More routes than exist

14. RouteID in Trips shows more routes than the Routes table contains.
15. Repair the formatting variants — delete nothing.
16. Report the distinct count before and after, and how many rows you repaired.

EVIDENCE

Before, after, and rows touched.

Rows before repair: 1200 rows

Rows after repair: 1200 rows

Rows repaired: 21 rows

DECISION

Written, in full sentences.

After repairing the rows, the same number of rows appeared. This only means that those formatting errors in the RouteID meant nothing more than just formatting errors — they were not duplicate records.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M Drawn at: 11:17AM Finished: 11:30AM

BAND B · TASK B3 · 10 POINTS

Two steps where the order matters

What the duplicates were worth

17. Remove duplicate rows from Tickets.
18. Report the row count before and after.
19. Report total FareAmount before and after, and the difference.

EVIDENCE

Both row counts, both revenue figures, and the peso difference.

Rows Before: 18329

Rows After: 18298

FareAmount Before: 2,528,647.94

FareAmount After: 2,523,700.41

FareAmount Difference: 4,947.53

DECISION

Written, in full sentences.

This data shows that there was a total of 4,947.53 in duplicate sales record, this claim is supported by the total number of remove duplicate rows of 31 rows in total.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M Drawn at: 11:30AM Finished: 11:35AM

BAND C · TASK C1 · 12 POINTS

A diagnostic — find it, count it, say what it costs

A bus that is not in the fleet

20. Using a left anti join, find every trip whose BusID is not in the Buses table.
21. Count them and name the value.
22. State what happens to those trips in a report grouped by bus type.

EVIDENCE

Count, value, and the silent consequence.

Unknown data rows: 17 rows

Value name: BU099

This distinct value represents a ghost data whenever reported blindly with further investigation.

DECISION

Written, in full sentences.

Investigate further as this might only be a formatting error.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M Drawn at: 11:36AM Finished: 11:37AM

BAND C · TASK C2 · 12 POINTS

A diagnostic — find it, count it, say what it costs

A ticket for a trip that never ran

23. Using a left anti join, find every ticket whose TripID is not in the Trips table.
24. Count them and name the value.
25. State what it means, in business terms, that these tickets exist.

EVIDENCE

Count, value, and what it implies about the business process.

Unknown data rows: 23 rows

Value name: T99999

This value could be a formatting error.

DECISION

Written, in full sentences.

Investigate further as this might only be a formatting error.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M Drawn at: 11:38AM Finished: 11:48AM

BAND C · TASK C3 · 12 POINTS

A diagnostic — find it, count it, say what it costs

The people nobody scheduled

26. Compare the Crew table against the DriverID and ConductorID columns in Trips.
27. Count the crew members who never appear on any trip.
28. State what those crew members have in common.

EVIDENCE

The count, and the pattern.

All the Crew for DriverID and ConductorID are present in the data and have appeared on the trips.

The missing values are the “Inspector” role.

Distinct values missing: 6 rows

DECISION

Written, in full sentences.

These crew members are not assigned on one particular trip.

Applied steps renamed? YES / NO Query name: _____

BAND D · TASK D1 · 15 POINTS

A judgement — decide, and defend it in writing

The trips that look wrong

- 29. Run an outlier test on FuelLitres in Trips.
- 30. Report the quartiles and the upper fence, and how many trips exceed it.
- 31. Decide: fix, exclude, escalate or accept. Defend it in writing.

EVIDENCE

Q1, Q3, the fence, the count beyond it, and your written decision.

DECISION

Written, in full sentences.

Name: Lagdamen, Lawrence M Drawn at: 11:57AM Finished: _____

BAND D · TASK D2 · 15 POINTS

A judgement — decide, and defend it in writing

Fact or dimension?

32. Open BayanihanBus_Maintenance.xlsx.
33. Decide whether it is a fact table or a dimension table.
34. If it is a fact, declare its grain as a sentence. If a dimension, name the fact it serves.
35. Defend the decision in writing.

EVIDENCE

Your classification, the grain sentence or join key, and the defence.

This is a dimension table.

It serves the fact table, Trips.

DECISION

Written, in full sentences.

This table is related to the table Trips, because it highlights the buses that aren't operational on a particular day because they are being maintained accordingly.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M Drawn at: 11:59AM Finished: _____

BAND D · TASK D3 · 15 POINTS

A judgement — decide, and defend it in writing

Should these two tables be joined?

- 36. Tickets carries TripID. Trips has one row per TripID.
- 37. Decide whether you would create a direct relationship from Tickets to Trips.
- 38. Defend the decision in writing, naming what you would do instead if you decline.

EVIDENCE

Your decision and its defence.

They have direct relation since a row in Tickets table corresponds to a row in Trips table

I would use the TripID from the Trips table to correctly identify the total sales of the certain TripID and the total number of tickets issued.

DECISION

Written, in full sentences.

This would correctly identify the total number of sales made by every trip conducted.

Applied steps renamed? YES / NO Query name: _____